

IMPACT STATEMENT

Latent Probing for Toxicity Detection

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Cohort: AI Saturdays Lagos - Cohort 10
Competition: Latent Probe Challenge - Toxicity Detection (Codabench)
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Purpose

Team OKU's project trains a linear probe - a lightweight classifier - on the internal representations of the Gemma 2-2B language model to detect harmful or toxic requests before the model generates a response. The project addresses a specific problem: safety infrastructure for AI systems is expensive, requiring either a second guardrail model or a commercial moderation API charged per message. This cost is prohibitive for African developers, universities, NGOs and public institutions building AI tools for their own communities - many of whom serve users writing in Nigerian Pidgin, Hausa, Igbo, or Yoruba.

Positive Impacts

The most significant positive impact is making safety screening accessible to builders who currently cannot afford it. Because our probe reuses hidden-state activations the model already computes, the additional cost of adding safety is negligible. The probe's design is also inspectable - a developer can examine which inputs trigger a flag and why, rather than accepting an opaque verdict from a foreign API. I want to be direct about scope: the probe currently trained and validated is English-only. The impact described above is delivered today for English-language deployments, and remains an intended future benefit - not yet a realised one - for the Pidgin, Hausa, Igbo, and Yoruba-speaking communities our problem statement is ultimately about.

Potential Risks and Harms

The most serious risk is dual use: a map of where safety-relevant features are encoded inside a model is also a map for evading them. A second risk is proxy discrimination - the probe may flag text mentioning specific ethnic groups, religions, or dialects rather than genuine harmful intent. A third, sharpened by our current English-only scope, is deployment mismatch: a builder who applies our probe to non-English traffic assuming it works as intended would be trusting an entirely unvalidated tool.

Values

Our project is guided by equity, accessibility, transparency, and proportionality. Transparency, in particular, requires us to state plainly where we have not yet delivered on our own stated goals, rather than describe the project as further along than it is.

Mitigation Strategies

We publish methodology and scope limitations clearly, including the English-only status of the current probe, so a deployer can judge for themselves where it is and is not appropriate to use. We are building a Stakeholder Engagement Plan centred on the communities our probe does not yet serve, as the concrete next step toward closing this gap rather than leaving it unaddressed.

Limitations and Uncertainties

Our training and evaluation data is entirely English. Planned translation work in Hausa, Igbo, and Nigerian Pidgin, with Yoruba held out as a documented gap, was not completed during the development phase; development time went instead to resolving data-labelling and extraction bugs discovered during testing. We have no validated performance data for any of the four languages named in our own problem statement. This is the central limitation of the current submission.